

CS & IT ENGINEERING



Computer Network

Routing

Lecture No. – 04

By - Abhishek Sir





Recap of Previous Lecture



Topic

Distance Vector Routing





Topics to be Covered



Topic

Distance Vector Routing

Topic

Routing Protocols

ABOUT ME



Hello, I'm **Abhishek**

- GATE CS AIR - 96
- M.Tech (CS) - IIT Kharagpur
- 12 years of GATE CS teaching experience

Telegram Link : https://t.me/abhisheksirCS_PW

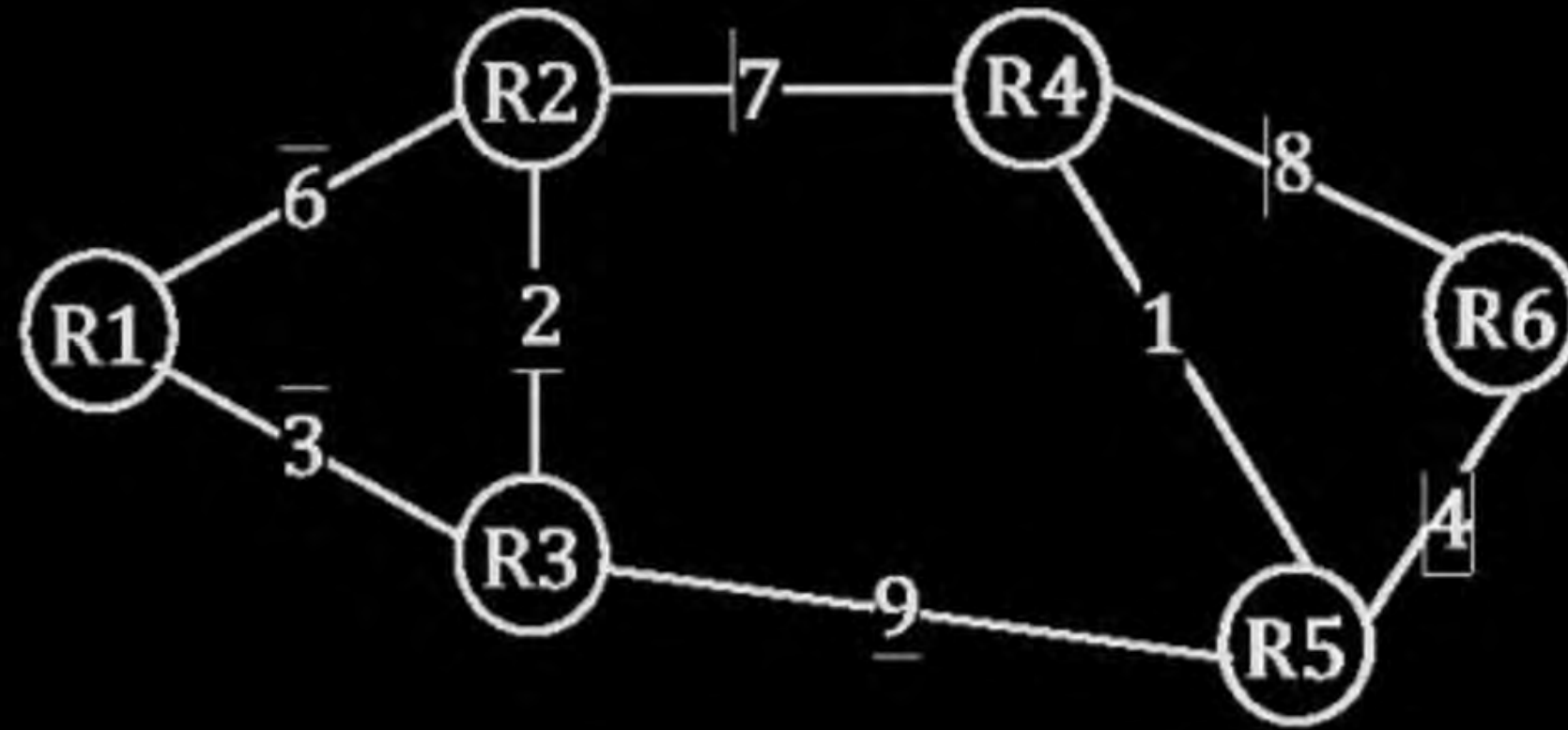


[MCQ]

[GATE-2010][2 Mark]



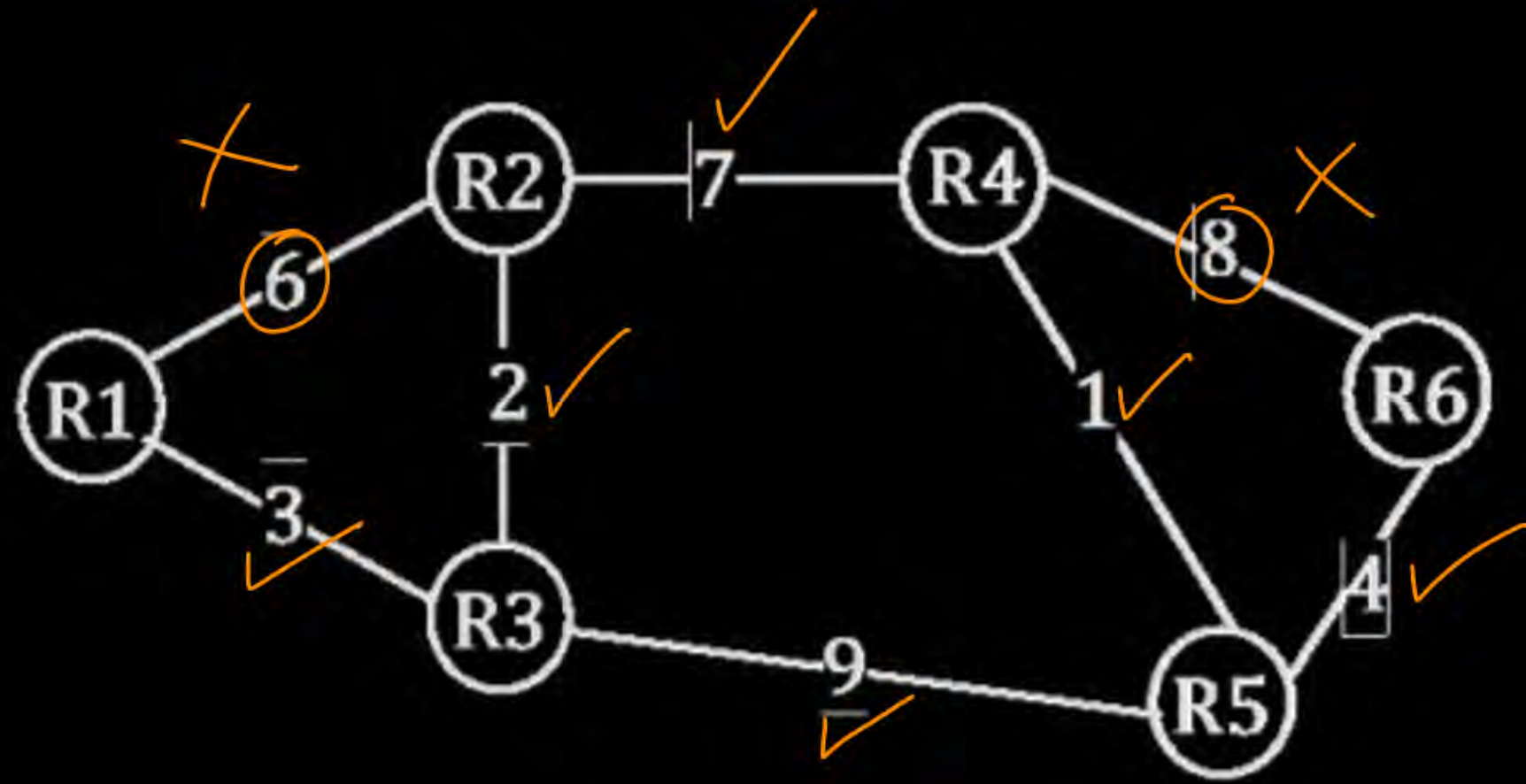
#Q. Consider a network with 6 routers (R1 to R6) connected with links having weights as shown in the following diagram :



All the routers use the (distance vector) based routing algorithm to update their routing tables. Each router starts with its routing table initialized to contain an entry for each neighbor with the weight of the respective connecting link. After all the routing tables stabilize, how many links in the network will never be used for carrying any data?

- A 1
- ☒ B 2
- C 3
- D 4

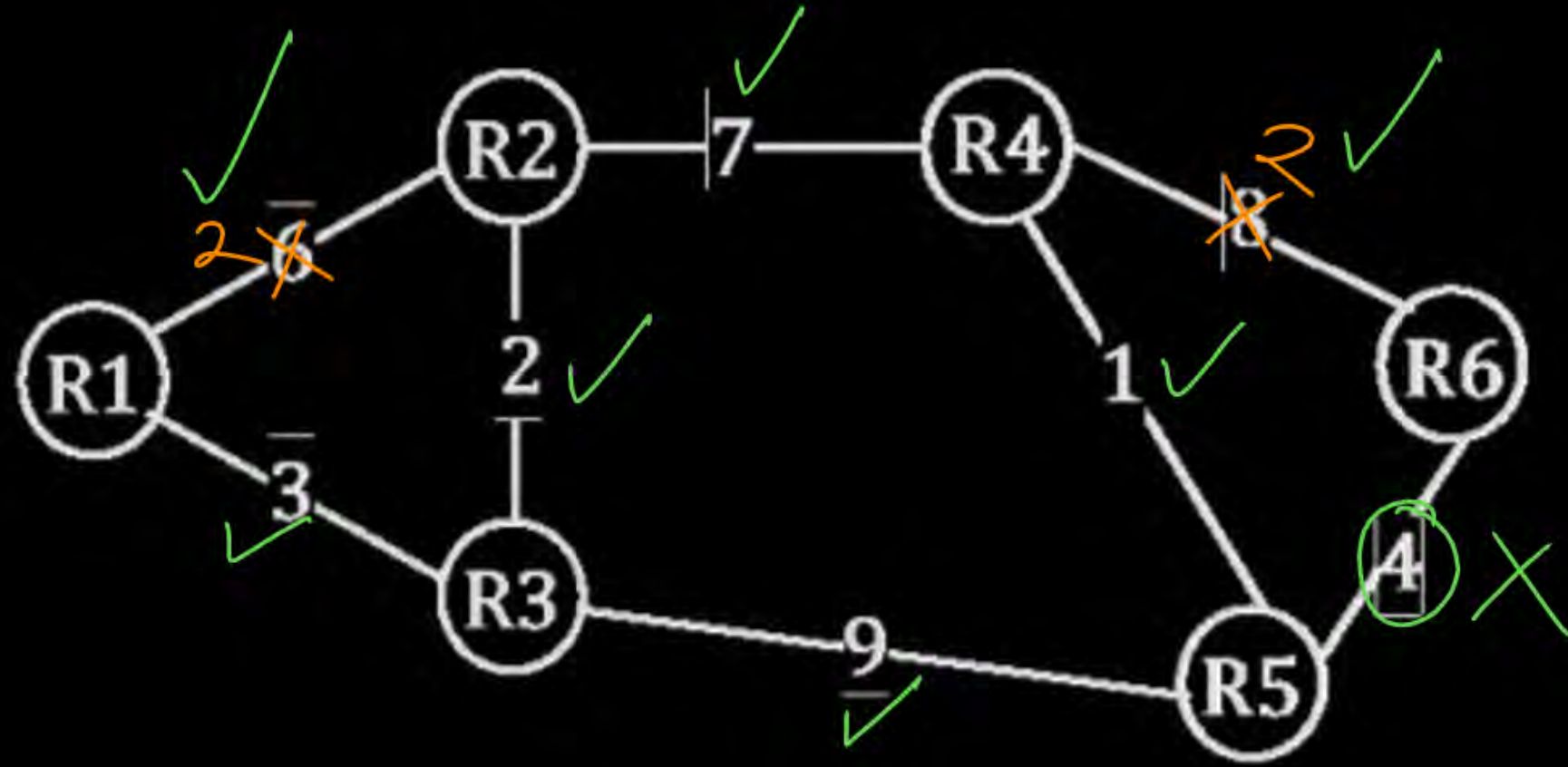
Ans: B



#Q. Suppose the weights of all unused links in the previous question are changed to 2 and the distance vector algorithm is used again until all routing tables stabilize. How many links will now remain unused?

- ☐ A 0
- ☒ B 1
- ☐ C 2
- ☐ D 3

Ans: B





Topic : Distance Vector Routing



→ Problem : May suffer with "Count to infinity" problem
[Link to infinity problem, occur due to routing loop in DVR]

CASE II (First Round):-



Node	Cost	Via
A	0	A
B	∞	
C	∞	

Node	Cost	Via
A	∞	
<u>B</u>	<u>0</u>	B
C	1	C

Node	Cost	Via
A	<u>2</u>	B
B	1	B
C	0	C

A	<u>∞</u>
B	0
C	1

A	<u>2</u>
B	1
C	0

CASE II (Second Round):-



Node	Cost	Via
A	0	A
B	∞	
C	∞	

Node	Cost	Via
A	3	C
B	0	B
C	1	C

Node	Cost	Via
A	∞	
B	1	B
C	0	C

A	3
B	0
C	1

A	∞
B	1
C	0

CASE II (Third Round):-



Node	Cost	Via
A	0	A
B	∞	
C	∞	

Node	Cost	Via
A	∞	
B	0	B
C	1	C

Node	Cost	Via
A	4	<u>B</u>
B	1	B
C	0	C

A	∞
B	0
C	1

A	4
B	1
C	0

[MCQ]

[GATE-IT-2005] [1 Mark]



#Q. Count to infinity is a problem associated with :

- ☒ A Link state routing protocol
- ☐ B Distance vector routing protocol
- ☒ C DNS while resolving host name
- ☒ D TCP for congestion control

Ans: B

[NAT]

(H.W.) IIT-KGP

[GATE-2022][1 Mark]



#Q. Consider a network with three routers P, Q, R shown in the figure below. All the links have cost of unity.



The routers exchange distance vector routing information and have converged on the routing tables, after which the link Q-R fails. Assume that P and Q send out routing updates at random times, each at the same average rate. The probability of a routing loop formation (rounded off to one decimal place) between P and Q, leading to count-to-infinity problem, is _____.



Topic : Distance Vector Routing



→ Problem : May suffer with “Count to infinity” problem
[Link to infinity problem, occur due to routing loop in DVR]

Solution 1 : Advertise routing path in ‘Distance Vector’
[Send ‘Next Hop’ (via) with distance vector]



Node	Cost	Via
A	0	A
B	1	B
C	2	B

Node	Cost	Via
A	1	A
B	0	B
C	1	C

Node	Cost	Via
A	2	B
B	1	B
C	0	C

Node	Cost	Via
A	0	A
B	1	B
C	2	B

Node	Cost	Via
A	1	A
B	0	B
C	1	C

Node	Cost	Via
A	1	A
B	0	B
C	1	C

Node	Cost	Via
A	2	B
B	1	B
C	0	C



Topic : Distance Vector Routing



- **Problem** : May suffer with “**Count to infinity**” problem
[**Link to infinity** problem, occur due to routing loop in DVR]

Solution 2 : “**Split horizon hack**”
[**Do not advertise** the **distance to a neighbor**,
if the **neighbor is the Next Hop** for **corresponding entry**]



Node	Cost	Via
A	0	A
B	1	B
C	2	B

Node	Cost	Via
A	1	A
B	0	B
C	1	C

Node	Cost	Via
A	2	B
B	1	B
C	0	C

A	0
---	---



B	0
C	1

A	1
B	0



C	0
---	---



Topic : Distance Vector Routing



→ **Problem** : May suffer with “**Count to infinity**” problem
[**Link to infinity** problem, occur due to routing loop in DVR]

Solution 3 : “**Poison reverse**”

[**Advertise infinite distance** to a **neighbor**,
if the **neighbor is the Next Hop** for **corresponding entry**]



Node	Cost	Via
A	0	A
B	1	B
C	2	B

Node	Cost	Via
A	1	A
B	0	B
C	1	C

Node	Cost	Via
A	2	B
B	1	B
C	0	C

A	0
B	∞
C	∞

A	∞
B	0
C	1

A	1
B	0
C	∞

A	∞
B	∞
C	0

CASE III :- [Poison Reverse]



Node	Cost	Via
A	0	A
B	∞	
C	∞	

Node	Cost	Via
A	∞	
B	0	B
C	<u>1</u>	C

Node	Cost	Via
A	<u>2</u>	<u>B</u>
B	1	B
C	0	C

A	<u>∞</u>
B	0
C	∞

A	<u>∞</u>
B	∞
C	0

CASE III :-



Node	Cost	Via
A	0	A
B	∞	
C	∞	

Node	Cost	Via
A	∞	
B	0	B
C	1	C

Node	Cost	Via
A	∞	
B	1	B
C	0	C



Topic : Distance Vector Routing



→ **Problem** : May suffer with “Count to infinity” problem
[Link to infinity problem, occur due to routing loop in DVR]

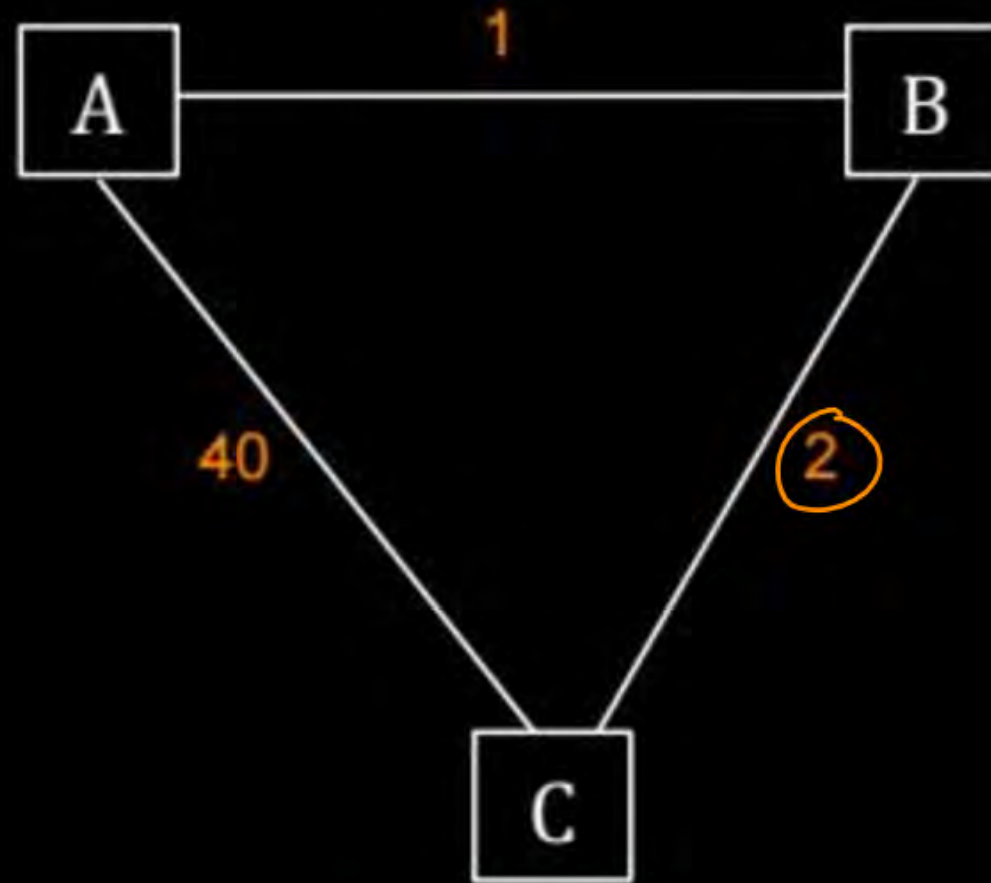
Solution 1 : Advertise routing path in ‘Distance Vector’
[Send ‘Next Hop’ (via) with distance vector]

Solution 2 : “Split horizon hack”
[Do not advertise the distance to a neighbor,
if the neighbor is the Next Hop for corresponding entry]

Solution 3 : “Poison reverse”
[Advertise infinite distance to a neighbor,
if the neighbor is the Next Hop for corresponding entry]

DV Routing (Without split horizon hack) :- [Stabilized Distance Vector]

Node	Cost	Via
<u>A</u>	<u>0</u>	<u>A</u>
<u>B</u>	<u>1</u>	<u>B</u>
<u>C</u>	<u>3</u>	<u>B</u>



Node	Cost	Via
<u>A</u>	<u>1</u>	<u>A</u>
<u>B</u>	<u>0</u>	<u>B</u>
<u>C</u>	<u>2</u>	<u>C</u>

Node	Cost	Via
<u>A</u>	<u>3</u>	<u>B</u>
<u>B</u>	<u>2</u>	<u>B</u>
<u>C</u>	<u>0</u>	<u>C</u>

Node	Cost	Via
A	1	A
B	0	B
C	20	C

Node	Cost	Via
A	3	B
B	20	B
C	0	C

19 B	21 B
21 B	5 B
21 B	21 B
21 B	7 B
21 B	21 B
21 B	⋮
	17 B
	21 B

4 A
20 C
6 A
20 C
8 A
⋮
20 C
18 A

20 C

20 C/A

20 C

20 C

20 C

	Router A	Router B
Link Cost Increases	$[C = 3, [\text{Via B}]]$	$[C = 20, [\text{Via C}]]$
$[\text{First iteration}]$	$C = 21, [\text{Via B}]$	$C = 4, [\text{Via A}]$
Second iteration	$C = 5, [\text{Via B}]$	$C = 20, [\text{Via C}]$
Third iteration	$C = 21, [\text{Via B}]$	$C = 6, [\text{Via A}]$
16 iteration	$C = 19, [\text{Via B}]$	$C = 20, [\text{Via C}]$
17 iteration	$C = 21, [\text{Via B}]$	$C = 20, [\text{Via C}]$

[MCQ]

IISC, H.W.

[GATE-IT-2008] [2 Mark]



#Q. Two popular routing algorithms are Distance Vector (DV) and Link State (LS) routing. Which of the following are true ?

- (S1) Count to infinity is a problem only with DV and not LS routing
- (S2) In LS, the shortest path algorithm is run only at one node
- (S3) In DV, the shortest path algorithm is run only at one node
- (S4) DV requires lesser number of network messages than LS

- A S1, S2 and S4 only
- B S1, S3 and S4 only
- C S2 and S3 only
- D S1 and S4 only

#Q. Consider the following three statements about link state and distance vector routing protocols, for a large network with 500 network nodes and 4000 links.

[S1] The computational overhead in link state protocols is higher than in distance vector protocols.

[S2] A distance vector protocol (with split horizon) avoids persistent routing loops, but not a link state protocol.

[S3] After a topology change, a link state protocol will converge faster than a distance vector protocol.

Which one of the following is correct about S1, S2, and S3 ?

- | | |
|---|---|
| ☐ A S1, S2, and S3 are all true | ☐ B S1, S2, and S3 are all false |
| ☐ C S1 and S2 are true, but S3 is false | ☐ D S1 and S3 are true, but S2 is false |

[MCQ]

IIT - M, H.W.

[GATE-2011][2 Mark]



#Q. Consider a network with five nodes, N1 to N5, as shown below. The network uses a Distance Vector Routing protocol. Once the routes have stabilized, the distance vectors at different nodes are as following.

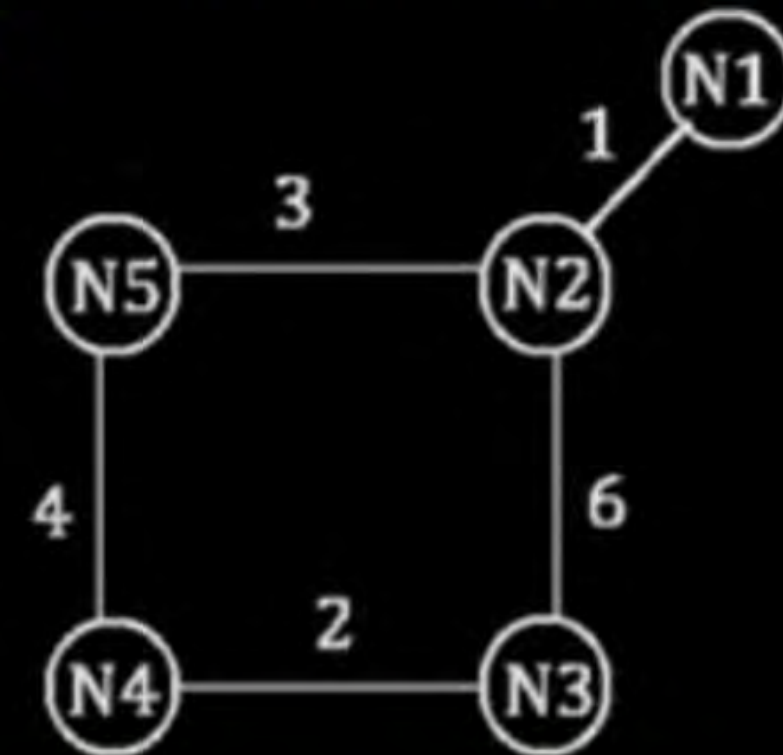
N1 : (0, 1, 7, 8, 4)

N2 : (1, 0, 6, 7, 3)

N3 : (7, 6, 0, 2, 6)

N4 : (8, 7, 2, 0, 4)

N5 : (4, 3, 6, 4, 0)



Each distance vector is the distance of the best known path at that instance to nodes, N1 to N5, where the distance to itself is 0. Also, all links are symmetric and the cost is identical in both directions. In each round, all nodes exchange their distance vectors with their respective neighbors. Then all nodes update their distance vectors.

In between two rounds, any change in cost of a link will cause the two incident nodes to change only that entry in their distance vectors.

The cost of link N2-N3 reduces to 2 (in both directions). After the next round of updates, what will be the new distance vector at node, N3?

- A (3, 2, 0, 2, 5)
- B (3, 2, 0, 2, 6)
- C (7, 2, 0, 2, 5)
- D (7, 2, 0, 2, 6)

[MCQ]

[H.W.]

[GATE-2011][2 Mark]



#Q. Consider the same data as given in previous question. After the update in the previous question, the link N1-N2 goes down. N2 will reflect this change immediately in its distance vector as cost, infinite. After the NEXT ROUND of update, what will be cost to N1 in the distance vector of N3 ?

- ☐ A 3
- ☐ B 9
- ☐ C 10
- ☐ D Infinite



Topic : OSPF

OSPF - LS - Dijkstra



→ Routing Protocol operate at Network Layer

→ OSPF flooding
→

→ Classic Link State Routing

→ OSPF packets are directly encapsulated into IP datagram payload field
[OSPF → IP]

→ Link cost metric : Bandwidth or delay

→ Each router has full topology
[Uses Dijkstra's algorithm to compute 'forwarding table']



Topic : RIP

[RIP-DV - Bellman Ford]



→ RIP : Routing Information Protocol

→ Routing Protocol operate at Network / Application Layer

→ Classic Distance Vector Routing

[Distance Vector exchanged every 30 second]

→ [Synchronous]
Message complexity
 $= O(n) \dots O(n^2)$

→ Based on the Bellman-Ford algorithm

→ RIP packets are directly encapsulated into UDP datagram payload field

[RIP → UDP → IP]

→ Link cost metric : Hop count

[MCQ]

(H.W.)

IIT-R

[GATE-2017] [1 Mark]



#Q. Consider the following statements about the routing protocols, Routing Information Protocol (RIP) and Open Shortest Path First (OSPF) in an IPv4 network.

- I. RIP uses distance vector routing
- II. RIP packets are sent using UDP
- III. OSPF packets are sent using TCP
- IV. OSPF operation is based on link-state routing

Which of the following above are CORRECT?

- A I and IV only
- B I, II and III only
- C I, II and IV only
- D II, III and IV only



Topic : BGP



→ BGP : Border Gateway Protocol

→ EGP : Exterior Gateway Protocol
[Inter-ISP routing]

→ Path Vector Routing
[Routing messages contains complete routes]

→ Routing Protocol operate at Application Layer

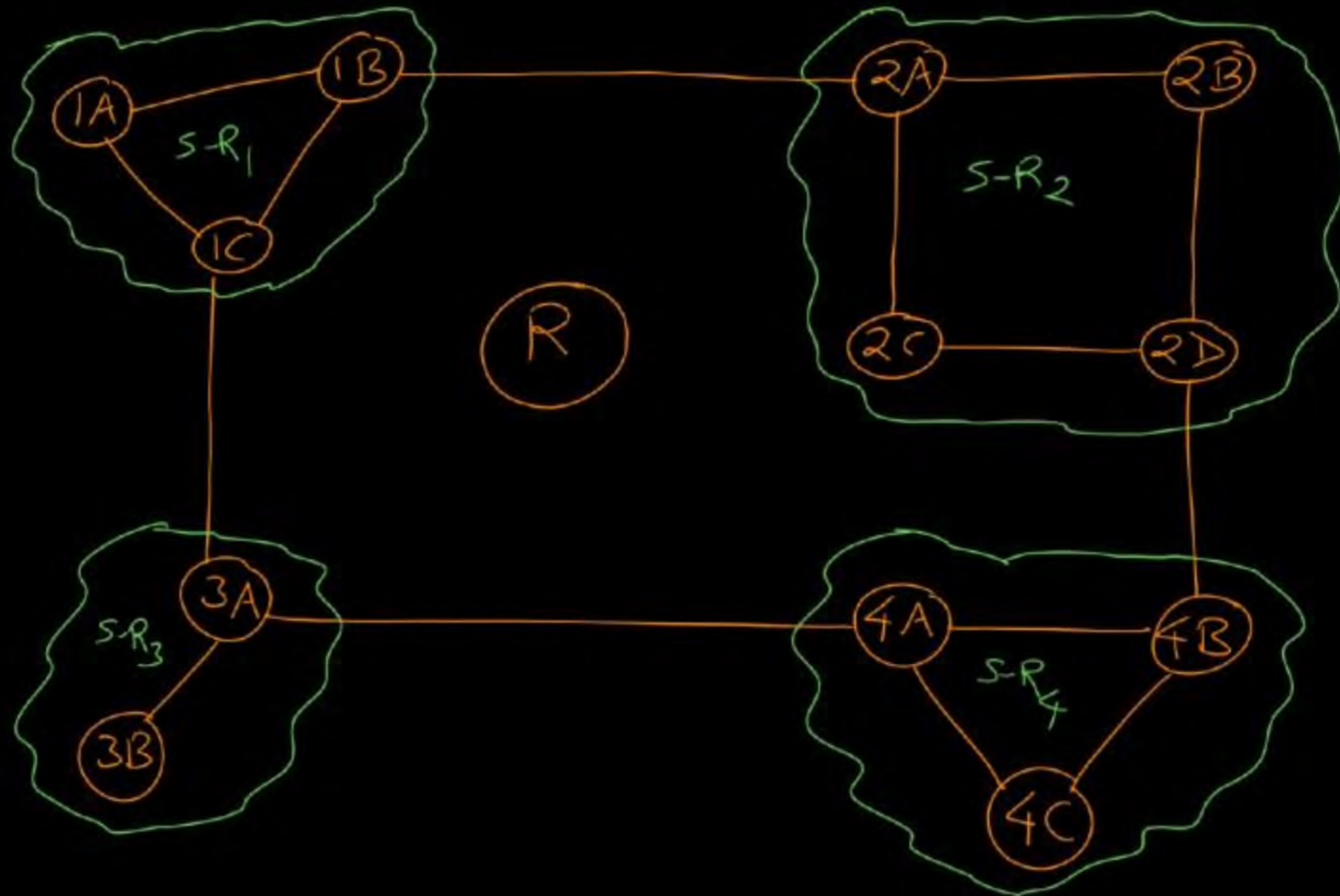
→ BGP packets are directly encapsulated into TCP segment payload field
[BGP → TCP → IP]



Topic : Hierarchical Routing



- The network is divided into regions and sub-regions.
[To create hierarchy]
- Allow efficient route management
- Route Aggregation
[To reduce the size of routing tables]



[Router 1A, global routing Table]

Destination	Next Hop
1A	1A
1B	1B
1C	1C
2A	1B
2B	1B
2C	1B
2D	1B
3A	1C
3B	1C
4A	1B/1C
4B	1B/1C
4C	1B/1C

Router 1A, Hierarchical Routing Table

Destination	Next Hop
1A	1A
1B	1B
1C	1C
2	1B
3	1C
4	1B/1C

[MSQ]

IIT-K, H.W.

[GATE-2023][1 Mark]



#Q. Which of the following statements is/are INCORRECT about the OSPF (Open Shortest Path First) routing protocol used in the Internet?

- ☐ A OSPF implements Bellman-Ford algorithm to find shortest paths.
- ☐ B OSPF uses Dijkstra's shortest path algorithm to implement least-cost path routing.
- ☐ C OSPF is used as an inter-domain routing protocol.
- ☐ D OSPF implements hierarchical routing.



Topic : Source Routing



→ Path addressing

→ Source host determine the route (partial or complete) for a packet
[before transmission]

→ Each packet carry determined routing path

[MCQ]

H.W.

IIT-D

[GATE-2020] [1 Mark]

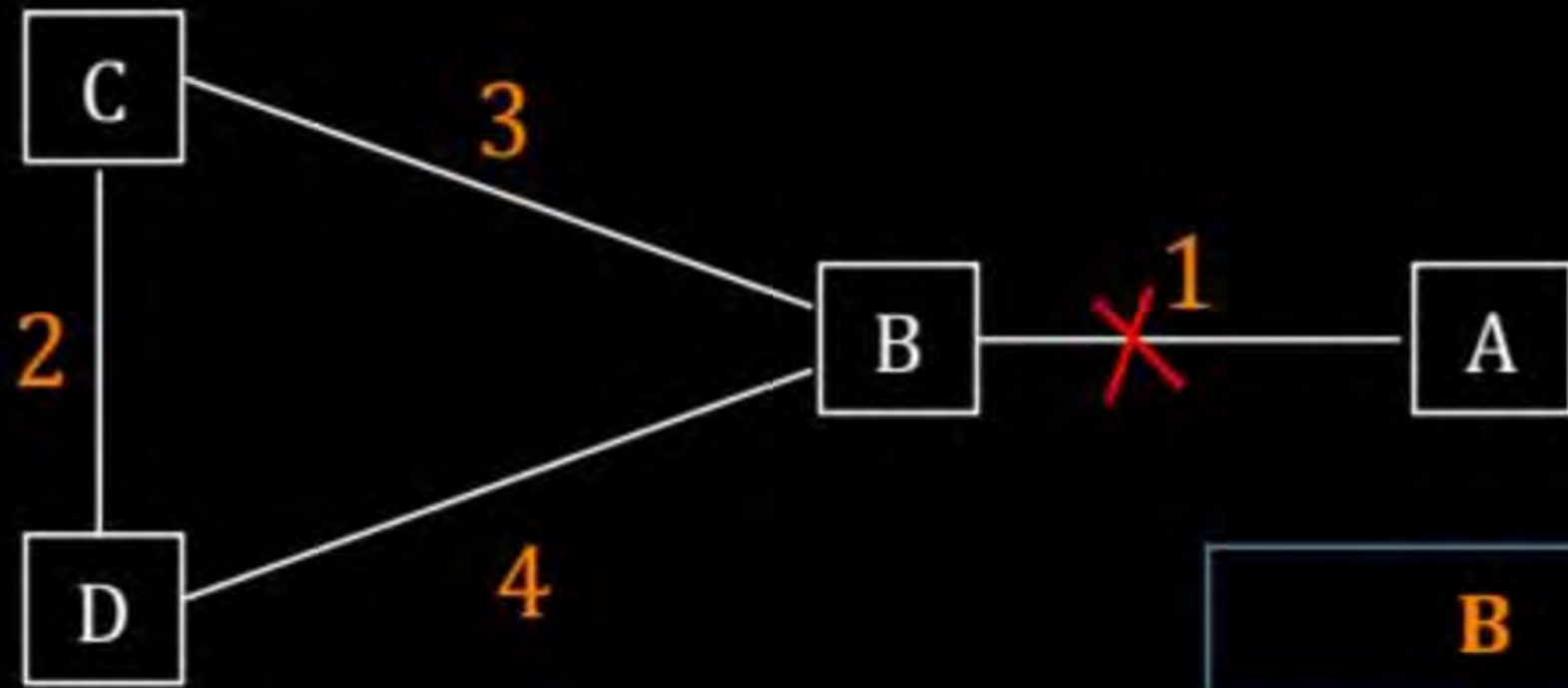


#Q. Consider the following statements about the functionality of an IP based router.

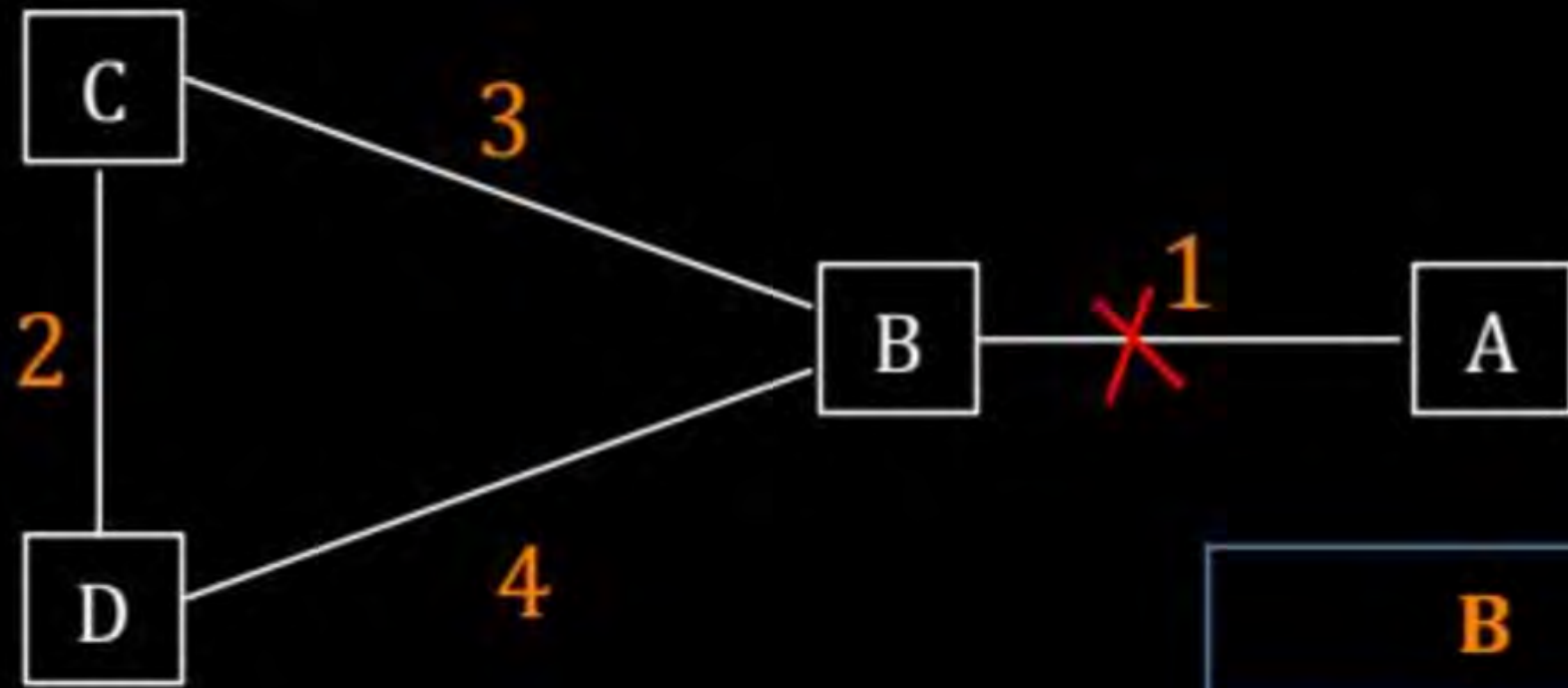
- I. A router does not modify the IP packets during forwarding.
- II. It is not necessary for a router to implement any routing protocol.
- III. A router should reassemble IP fragments if the MTU of the outgoing link is larger than the size of the incoming IP packet.

Which of the above statements is/are TRUE ?

- ☐ A I and II only
- ☐ B I only
- ☐ C II and III only
- ☐ D II only



B	C	D
A = 1, Via A	A = 4, Via B	A = 5, Via B
A = ∞	A = 4, Via B	A = 5, Via B



B	C	D
A = 1, Via A	A = 4, Via B	A = 5, Via B
A = ∞	A = 4, Via B	A = 5, Via B
A = ∞	A = 7, Via D	A = 6, Via C
A = 10, Via C	A = ∞	A = ∞
A = ∞	A = ∞	A = 14, Via B
A = ∞	A = 16, Via D	A = ∞



2 mins Summary



Topic

Distance Vector Routing

Topic

Routing Protocols



THANK - YOU

